YouTube + Facebook AI Automation Agent

Complete Professional System Structure & Workflow

End-to-end content automation system: trend research → idea generation → scoring → fact-checking → script → creative production → publishing → analytics → continuous optimization.

> **Core rule:** the automation should use authorized APIs/accounts and should not perform fake engagement, spam, unauthorized access, copied/copyright-infringing content, or platform-rule bypasses.

1. Complete System Architecture

Module	Main Job	Output
Trend Intelligence	Search trends, public content patterns, niche changes, own analytics	Trend records
Idea Engine	Generate and score content angles	Ranked ideas
Research Agent	Collect and verify facts/sources	Research brief
Script Agent	Hooks, structure, narration, CTA	Final script
Creative Director	Titles, thumbnails, storyboard, B-roll	Production brief
Production	Voice, visuals, captions, rendering	Video assets
Publisher	Metadata and scheduling	Published/scheduled post
Analytics	Performance and audience analysis	Performance report
Optimizer	Experiments and next ideas	Optimization actions

Every module passes structured records to the next module so every content item remains traceable from research to final result.

2. Trend & Research Engine

- YouTube/search trend signals and relevant queries.
- Public YouTube metadata and audience-facing topic patterns.
- Facebook public content where permitted by the platform/API.
- News/RSS/web sources for current topics, with dates and source authority.
- Competitor/channel watchlist: topics, formats, hooks and content gaps.
- Your own YouTube Analytics: audience, formats, traffic and retention.

Trend record: topic_id, niche, keyword, source_url, source_date, freshness, audience_fit, competition, content_gap, risk_level, recommended_format.

3. Idea Generator & Scoring

Factor	Weight	Question
Audience demand	25%	Does the audience have a real interest/problem?
Freshness	15%	Is the angle newly relevant?
Content gap	15%	Is there useful missing coverage?
Hook potential	15%	Can value/curiosity be clear immediately?
Feasibility	10%	Can it be produced well with available tools?
Discovery fit	10%	Does searchable language exist?
Originality/risk	10%	Is it original and low-risk?

Output: selected idea + alternative hooks + audience + format + research questions + risk flags.

4. Research & Fact-Checking Agent

- Prioritize primary and official sources.
- Store important claims with source and date.
- Flag conflicting information instead of silently combining it.
- Verify names, numbers, dates and quotations.
- Summarize source material rather than copying it.
- Block unsupported sensational claims or send them to human review.

Research object: claim → source → date → confidence → script_usage → verification_status.

5. Scriptwriting Engine

- Long-form: Hook → Promise → Context → Main points → Examples → Lesson → CTA.
- Short-form: Hook → Fast value → Curiosity → Payoff → CTA.
- Explainer: Problem → Cause → Evidence → Solution → Takeaway.
- Current topic: Verified fact → Context → What changed → Implications → Sources.
- Story: Setup → Tension → Reveal → Lesson.
- Facebook-native: shorter caption + discussion prompt.

Controls: duration, speaking speed, language, tone, audience, reading level, prohibited claims, CTA style and source requirements.

6. Creative Production Pipeline

- Scene-by-scene storyboard with timestamps.
- Voiceover text and pronunciation notes.
- B-roll/visual asset list.
- On-screen captions and emphasis words.
- Thumbnail concepts and title variants.
- YouTube description and chapters draft.
- Short-form cut points from long-form master.
- Asset/license ledger for stock, music, images and generated media.

7. Video Assembly & Quality Control

Orchestrator → TTS/voice → visual assets → subtitles → FFmpeg/render worker → QC → storage.

- Audio present and understandable.
- Captions synchronized.
- No missing media.
- Correct aspect ratio.
- Thumbnail/title accurately represent content.
- Copyright/license flags resolved.
- No accidental private information.
- Duration within requested range.

8. YouTube Automation

• Connect through authorized YouTube APIs/OAuth.

• Generate title, description, chapters and metadata.

• Prepare thumbnail and playlist settings.

• Schedule/publish after required approval gate.

• Store returned video ID and metadata snapshot.

• Collect Analytics after publication.

• Feed performance back into the optimizer.

9. Facebook Automation

- Use Meta/Facebook approved integrations and permissions.
- Generate Facebook-native captions from master content.
- Prepare and schedule videos/posts through authorized workflows.
- Store post ID, creative version and schedule status.
- Collect permitted performance metrics.
- Classify comments and route important questions to assistant/human.

Do not automate fake accounts, mass unsolicited messaging, artificial engagement, private-data scraping or rate-limit bypasses.

10. Analytics & Optimization Loop

Signal	Check	Optimization
Packaging	Impressions/CTR where available	Test title/thumbnail concepts
Retention	Drops, spikes, average view duration	Improve hooks/structure
Engagement	Comments, shares, likes where available	Extract questions/themes
Audience	New/casual/regular viewers	Adjust content mix
Traffic	Search/suggested/external	Improve discovery
Topics	Topic clusters over time	Shift future idea allocation

11. Community / Comment Agent

- Classify comments: question, praise, criticism, spam, lead, support or safety issue.
- Draft replies in brand voice.
- Escalate sensitive/legal/medical/financial issues to human review.
- Extract recurring questions into future content ideas.
- Build FAQ/topic clusters from genuine audience feedback.

12. Human Approval Gates

• Gate A — idea approval, optional.

• Gate B — research verification for high-risk/current claims.

• Gate C — script approval, configurable.

• Gate D — final video/thumbnail QC, recommended.

• Gate E — publishing, automatic only for trusted low-risk content.

• Gate F — weekly strategy review.

13. Technical Architecture

Component	Suggested Role
Frontend	Next.js / React
Backend	Node.js/Next.js API or Python FastAPI
Database	Supabase/PostgreSQL
Queue	Redis/BullMQ or equivalent
Storage	S3-compatible object storage
LLM	Provider abstraction for model switching
Research	Approved search/web/data APIs
Video	FFmpeg + approved TTS/media services
YouTube	YouTube Data + Analytics APIs
Facebook	Meta Graph API / approved integration
Auth	OAuth 2.0 + encrypted tokens
Monitoring	Logs, alerts, job status
Scheduler	Cron/queue worker with retries

14. Database Structure

• channels: platform, name, timezone, niche, brand_voice.

• research_items: source, topic, claims, date, verification_status.

• ideas: angle, score, format, status, selected_reason.

• scripts: version, language, duration, hook, body, CTA, approval.

• assets: type, source, license_status, file_url.

• videos: render_status, duration, master_url, thumbnail_url.

• publishes: platform, post_id, scheduled_at, status, metadata.

• analytics: post_id, date, views, retention, CTR where available, engagement.

• experiments: hypothesis, variable, baseline, result, decision.

• tasks: job_type, status, retries, error, next_run_at.

15. Master Automation Workflow

Scheduler → Trend Scan → Topic Clustering → Idea Generation → Idea Scoring → Research/Fact Check → Script → Quality Check → Approval → Storyboard/Creative → Voice/Visual Production → Render → QC → YouTube/Facebook Scheduling → Publish → Analytics Collection → Performance Diagnosis → Experiments → Next Ideas.

This creates a closed feedback loop: every published asset becomes input for future strategy decisions.

16. Admin Dashboard

• Overview: channels, jobs, scheduled posts, failures.

• Trend Radar: topics, freshness, sources, confidence.

• Idea Board: New → Research → Script → Review → Approved → Published.

• Script Studio: versions, sources, approvals.

• Production Board: render/asset status.

• Publishing Calendar: YouTube + Facebook.

• Analytics: video/topic performance.

• Experiment Center: hook/title/thumbnail tests.

• Automation Settings: schedules, models, limits, approval gates.

• Security: OAuth connections, audit log, kill switch.

17. Security, Copyright & Reliability

- Encrypt OAuth tokens; never expose secrets in frontend.
- Least-privilege permissions and revoke/disconnect controls.
- Rate limits, retries, idempotency and duplicate-post protection.
- Research timestamps and source records.
- Asset license/copyright ledger.
- Prompt-injection defenses for external content.
- Audit trail for generation/edit/publish actions.
- Global kill switch for publishing.
- Daily API/cost limits.
- Human review for sensitive claims.

18. Example Daily Cycle

06:00 Trend scan → 07:00 clustering → 08:00 ideas → 09:00 research → 10:00 scripts → 12:00 review → 13:00 production → 15:00 QC → 16:00 scheduling/publishing → evening analytics → nightly optimization report.

Timing should be configurable by channel and audience data.

19. Agent Roles / Prompt Modules

- Trend Analyst Agent
- Idea Strategist Agent
- Research / Fact Checker Agent
- Scriptwriter Agent
- Editor Agent
- Creative Director Agent
- Production Agent
- Publisher Agent
- Community Agent
- Analytics Agent
- Optimization Agent
- Compliance / QC Agent

Separate modules make debugging, cost control and quality improvement easier than one giant prompt.

20. Final Output Per Content Item

- Trend/research brief
- Idea + score
- 3–5 titles
- 3 hooks
- Final script
- Source/claim sheet
- Storyboard
- Voiceover
- Captions/SRT
- Visual/B-roll list
- Thumbnail concepts
- YouTube metadata
- Facebook caption
- Publishing schedule
- Published IDs/status
- Analytics snapshot
- Optimization report
- Next-content recommendations

Final result: a reusable content factory covering research, strategy, writing, production, publishing and learning, with high-risk actions behind approval controls.